Information System Management

Course No: 0612-301

Global E-Business

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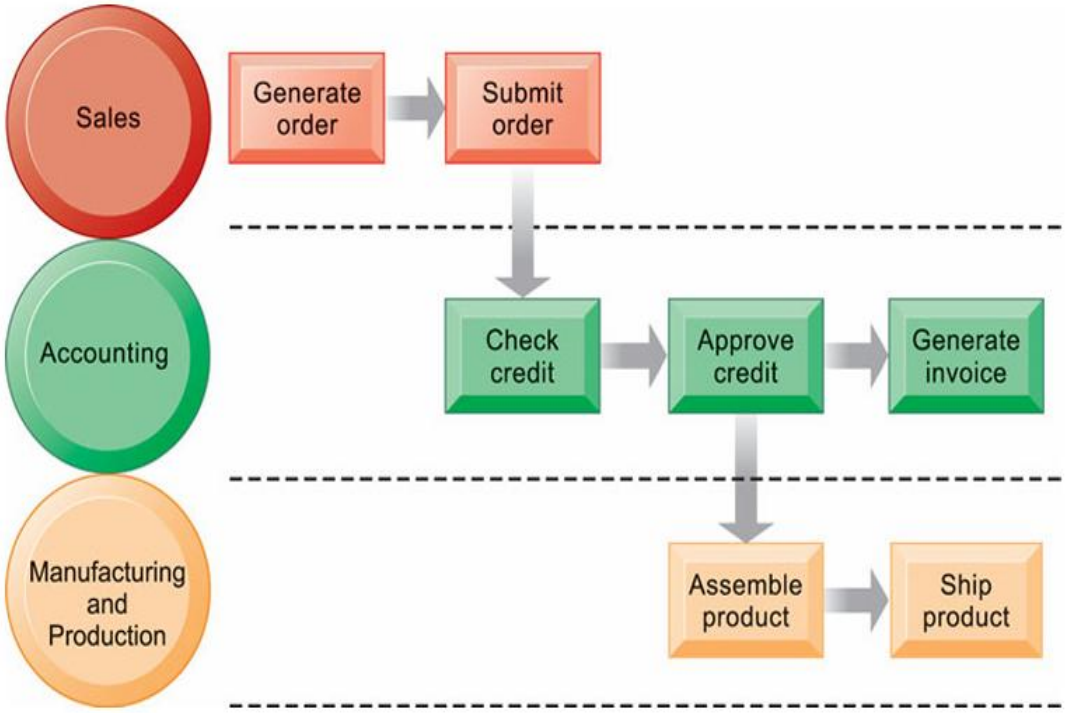
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Business System

- Every business can be seen as a collection of business processes.

FUNCTIONAL AREA	BUSINESS PROCESS
Manufacturing and production	Assembling the product Checking for quality Producing bills of materials
Sales and marketing	Identifying customers Making customers aware of the product Selling the product
Finance and accounting	Paying creditors Creating financial statements Managing cash accounts
Human resources	Hiring employees Evaluating employees' job performance Enrolling employees in benefits plans

Business System



- Fulfilling an order requires close co-ordination between major functional groups in a firm.
- The required information must flow rapidly –
  - Within the firm from one decision maker to another
  - With the business partners
  - With the customers

How IS affects Businesses?

- Automates many steps.
- Change the flow of the information.
- Allows more people to access and share the information.
- Replaces sequential steps with simultaneous tasks.
- Eliminates delays.
- Supports entirely new business models.

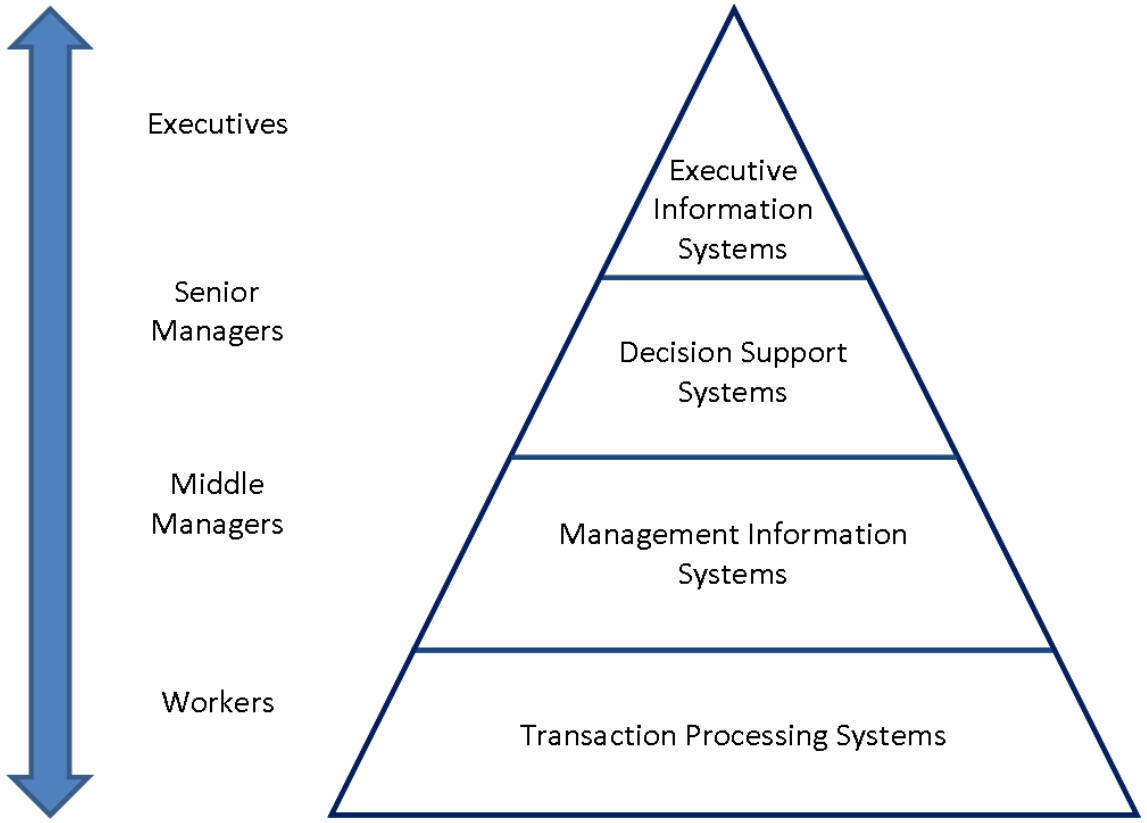
Electronic Business or E-Business

- E-business refers to the use of digital technology and the Internet to execute the major business processes in the enterprise.

- E-Business = business + digital information system

- Includes activities for –
  - Internal management for the firms.
  - Co-ordination with the suppliers and customers.
  - E-commerce – buying and selling goods and services over the Internet.

Information System in a Business



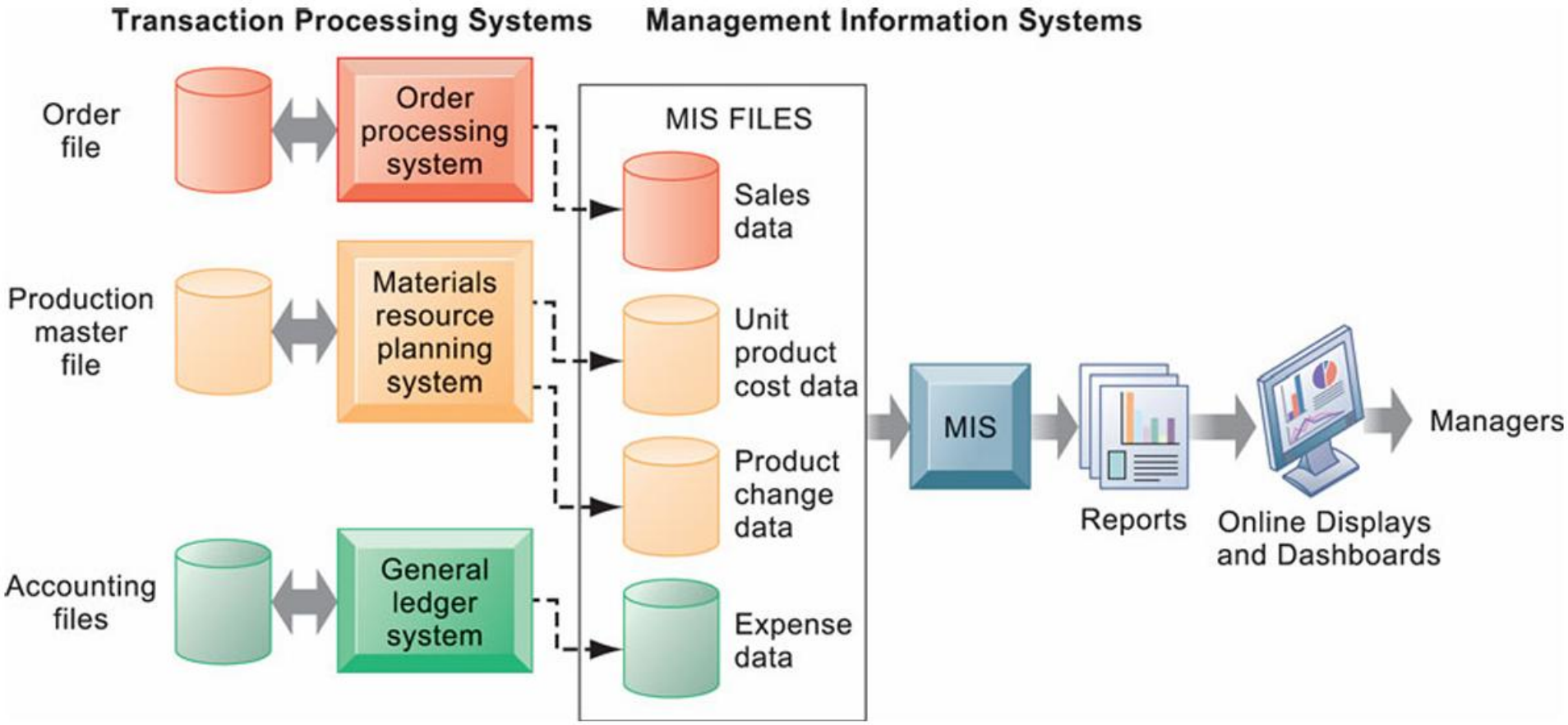
Transaction Processing Systems (TPS)

- A computerized system that performs and records the daily routine transactions necessary to conduct business.
- Examples: sales order entry, hotel reservations, payroll, employee record keeping, shipping, etc.
- Answers routine questions.
- Tracks the flow of transactions through the organizations.
- Used by operational managers and workers.
- TPS failure can cause heavy blow to the business.

Management Information System (MIS)

- Serves middle management.
- Provides summarized reports on the organizations' current performance regularly.
- This information is used to monitor and control the business and predict future performance.
- Provides answer to routine questions according to a predefined procedure instead of sophisticated mathematical models or statistical techniques.

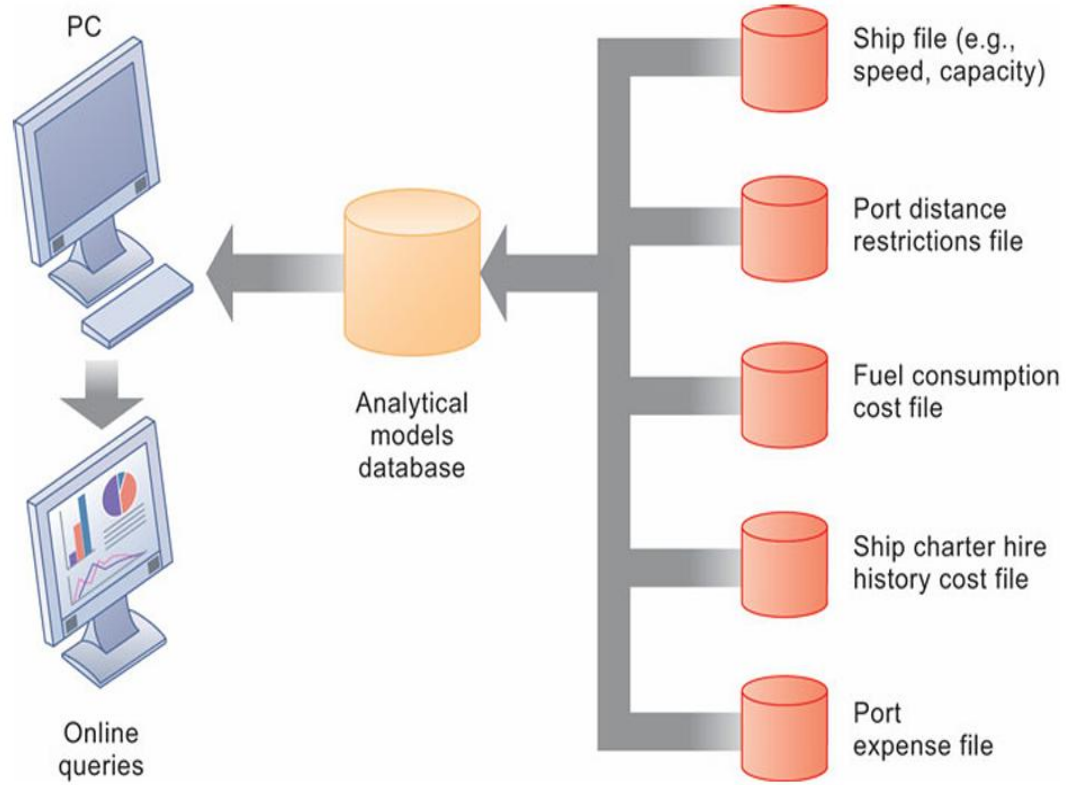
Management Information System (MIS)



Decision Support Systems (DSS)

- Focus on problems that are unique and rapidly changing.
- Try to arrive to a solution that may not be defined before.
- Designed to support middle and all upper managements.
- Often bring in information from external sources.
- Use many mathematical, statistical, or logical models to analyze.

Decision Support Systems (DSS)



A voyage estimating system for a firm that performs shipping contracts

- Which vessel should be assigned at what rate to maximize profits?
- What is the optimal speed at which a particular vessel can maximize its profit and still meet its delivery schedule?
- What is the optimal loading pattern for a ship bound for the U.S. West Coast from Malaysia?

Information System in a Business

- Business intelligence is the tools that –
  - Organizes, analyzes, and provides access to data.
  - Helps managers and others to make more informed decisions.
  - Can be found at all levels of the organizations.
  - Includes both MIS and DSS.

- Executive support systems (ESS) –
  - Help senior management make decisions.
  - Address non-routine decisions requiring judgment, evaluation, and insight because there is no agreed-on procedure for arriving at a solution.
  - Use graphs, charts, and others.
  - Incorporate external data and summarized internal data from MIS and DSS.
  - Filter, compress, and track critical data.

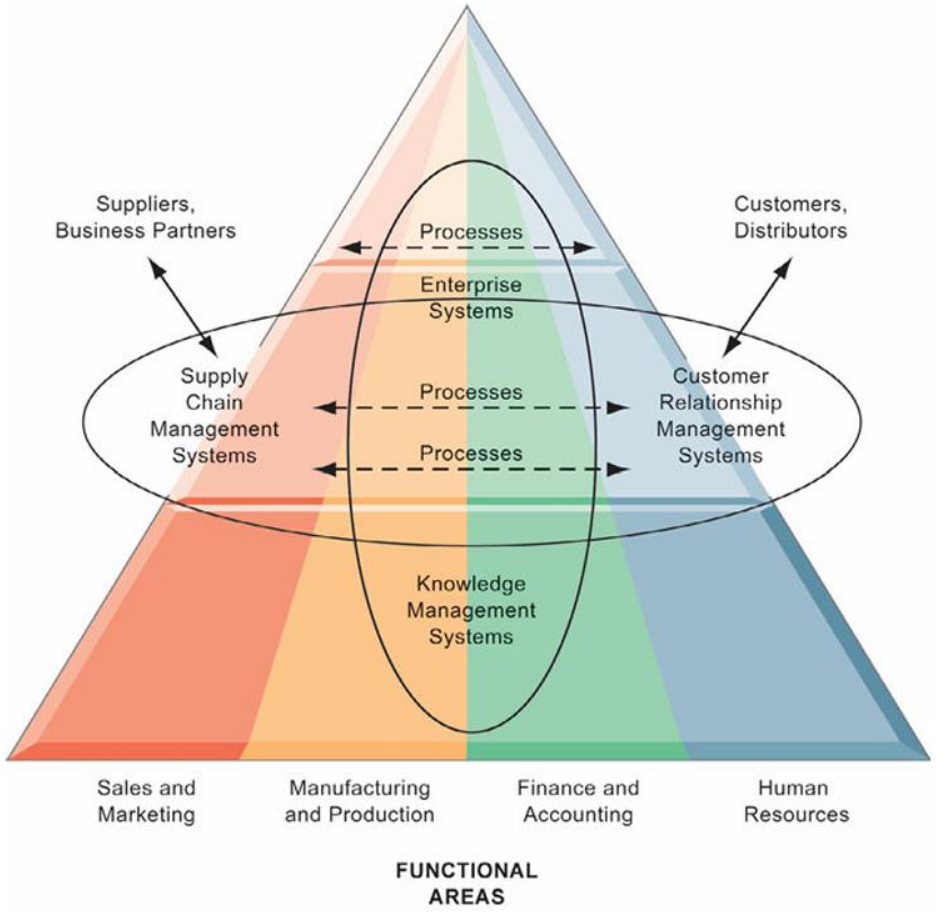
Enterprise Applications

- So, how to make all these work together?

- Enterprise applications include all levels of management and span all functional areas.

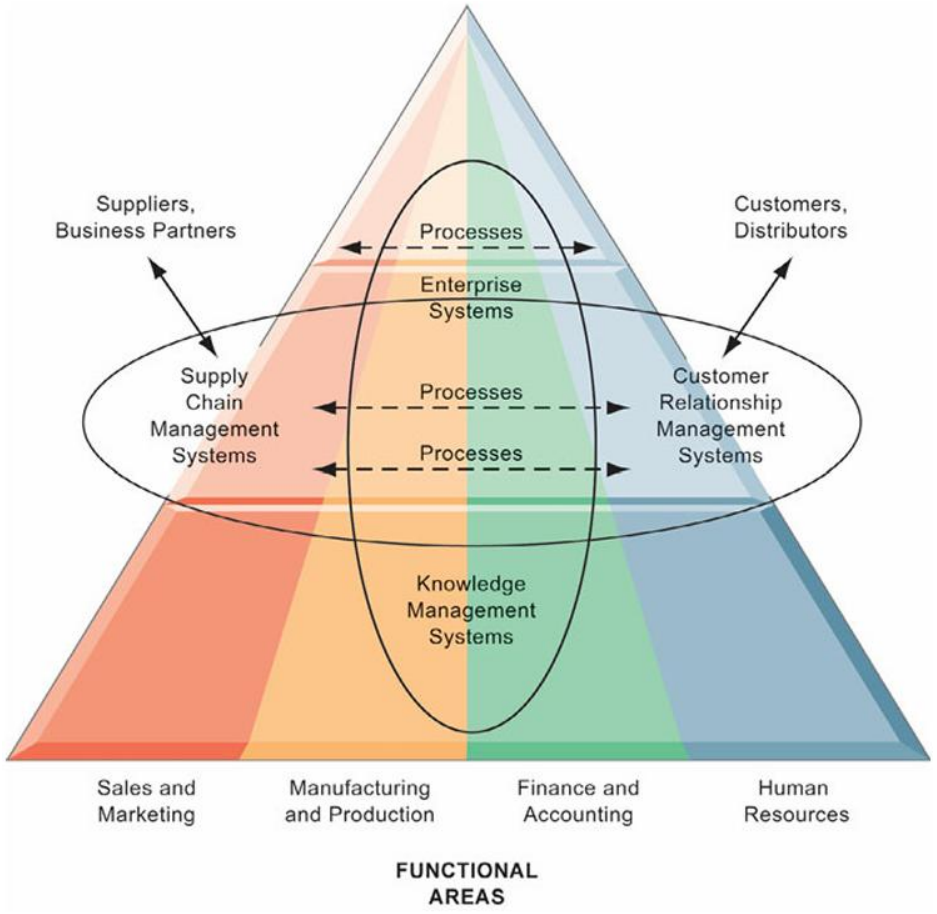
- Four types –
  - Enterprise systems (a.k.a. ERP – Enterprise Resource Planning)
  - Supply chain management systems (SCM)
  - Customer relationship management systems (CRM)
  - Knowledge management systems (KMS)

Enterprise Applications



- ERPs –
  - integrate business processes in all functional areas in a single software system.
- SCMs –
  - get the right amount of product in the least time and lowest cost.
  - manage relationship with suppliers.
  - share information between suppliers, purchasing firms, distributors, and logistics companies.
  - automate the flow of information across organizational boundaries.

Enterprise Applications



- CRMs –
  - manage relationship with customers.
  - sales, marketing, customer services, etc.
  - identify, attract, and retain the most profitable customers.
  - provide better service to existing customers.
  - increase sales.

- KMSs –
  - collect all relevant knowledge and experience in the firm.
  - improve business processes and management decisions.

Collaboration and Teamwork

- Working with others to achieve shared and explicit goals.

- Benefits –
  - Increased productivity.
  - Better quality.
  - Innovation.
  - Better customer service.
  - Financial growth.

- Use different modern tools to collaborate.

The Information Systems Department

- Programmers
- System analysts
- Managers
- CIO, CSO, CKO, CPO
- End users

References

- System Analysis and Design – Elias M. Awad
  - Chapter 1
- Management Information System – Kenneth C. Laudon, Jane P. Laudon
  - Chapter 2
- Internet

The END